

Fundamentals of Real Estate Analysis

MGTF 408 Real Estate Finance

Spring 2025

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DESCRIPTION

The course will draw on the principals from corporate finance and investments and apply them to the context of real estate markets in their various shapes and forms (private vs. public markets; debt vs. equity). However, we may at times make reference to the complementary fields of real estate law, public and urban economics, design and architecture, and the economics of inequality.

OBJECTIVES

While the study of real estate can be undertaken from many different perspectives, we will choose the value perspective as our major theme. What is the value of a particular asset? This question is central to all real estate decision making: acquiring a property, mortgage underwriting, real estate development, lease vs. buy decisions, to name a few. Additionally, the course will focus on commercial real estate (CRE) assets. Office, Multi-family Residential, Retail and Industrial are the four main types of commercial real estate. The CRE landscape also encompasses assets such as hotels, leisure facilities, data centers, cell-phone towers, among others. Finally, we will address historical trends in real estate, including the legacy of residential segregation, redlining, and reverse-redlining.

Fundamentals of Real Estate Analysis thus has the following objectives:

To develop students' awareness of a fundamental value-based framework—both its strengths and weaknesses—from which to analyze a key set of real estate decisions.

To heighten students' awareness of macroeconomic and local demand and supply factors which drive value.

To increase students literacy of the historical drivers of urban patterns and the role of public institutions in driving or correcting historical patterns.

To introduce students to financial modelling techniques in real estate investment analysis. To understand the impact of critical assumptions (model inputs) on valuation and risk metrics (model outputs). Consequently, to make students aware of the strengths and weaknesses of these widely used techniques, and help them develop an intuition for, and a critical eye towards, these models.

To develop an understanding of risk and its relationship to return in the context of real estate investments. Additionally, to understand how risk interacts with the financing and capital structure of real estate investments.

MATERIALS

Required

- Text: Geltner, Miller, Clayton and Eichholtz Commercial Real Estate Analysis and Investments, 3rd Edition.
- Rothstein, Richard, The color of law: a forgotten history of how our government segregated America, 1st Edition.

Recommended

- Schuetz, Jenny, Fixer-Upper: How to Repair America's Broken Housing Systems.
- Additional Readings: The course will rely heavily on additional readings of academic papers, white papers, and news articles.

SCHEDULE

- I. **Weeks 1-6: Building Valuation Framework – ~2 hours in-person lecture**
- II. **Weeks 1-6: Understanding Real Estate Trends – ~50 min asynchronous lecture**
- III. **Week 7: Exam**
- IV. **Weeks 9-10: Understanding Urban Economics – 2:50 hours in person**

Class	Topic	Comments
Week 1	Course Introduction/Historical Trends in Real Estate Markets	Chapter 2.1 and Chapter 2.2 of Geltner
Week 1	Space Market and the Asset Market	Chapter 1
Week 1	Asynchronous: Topics in Real Estate (Housing Affordability)	Online Recording 1
Week 2	Measuring Returns and Introduction to Price Indices	Chapter 9, NY Fed short article.
Week 2	Market Analysis	Chapter 6
Week 2	Asynchronous: Topics in Real Estate (Returns and Market Analysis with Climate Change)	Online Recording 2
Week 3	Leasing Strategy I: Fundamentals	Chapter 30
Week 3	Leasing Strategy II: Building a Pro Forma	
Week 3	Asynchronous: Topics in Real Estate (The Short-Term Rental Boom)	Online Recording 3
Week 4	Valuation I	Chapter 10
Week 4	Valuation II	Chapter 11
Week 4	Asynchronous: Topics in Real Estate (Prices, Neighborhood Revitalization, and Gentrification)	Online Recording 4
Week 5	Mortgage Basics and the Effect of Leverage on Expected Returns	Geltner Chapter 13 Chapter 16.1-16.3;

		Chapter 17.1; Chapter 17.2.1
Week 5	Incorporating Debt into the Pro-Forma	
Week 5	Asynchronous: Topics in Real Estate (The Great Financial Crisis of 2008)	Online Recording 5
Week 6	Real Estate Investment Capital Structure	Chapter 15
Week 6	Real Estate Investment Trusts	Chapter 23
Week 6	Asynchronous: Topics in Real Estate (How Much Institutional Investors Affect Neighborhoods?)	Online Recording 6
Week 7	Midterm Exam	
Week 8	Dueling Markets	
Week 8	Urban Economics I: Fundamentals/ Presentations	Rothstein/Schuetz
Week 9	Urban Economics II: Inequality/ Presentations	Rothstein/Schuetz
Week 10	Urban Economics III: Place Based Policies and Public Finance/ Presentations	Rothstein/Schuetz

Exams

- You will be required to take one exam, which will be closed-book closed-notes (No computers or programmable calculators are allowed)
- You will be provided with a sheet containing the important formulae
- You can bring a financial calculator
- Exam is mandatory. Exceptions are given only for family or medical emergencies, in which case prior notice and hard evidence are required. If you have to miss a mid-term exam because of these emergencies, then the grade portion of that exam will be cumulated to the grade portion of your final presentation.
- I do not host make up exams for the midterm exam.
- No lectures or office hours are given on the exam dates.

Presentation and Write-Up

You will be required to give a team presentation and provide a four-page write-up based on a topic from “The color of law: a forgotten history of how our government segregated America” or “Fixer-Upper: How to Repair America’s Broken Housing Systems.” Attendance, or lack thereof, and participation during your colleagues’ presentations will be taken into account in the grading for up to 25% of the presentation grade.

Class Participation

You are strongly encouraged to participate in this class. These participations can be in the form of raising questions, providing answers to questions, or bringing related outside materials to the attention of the class.

I will record attendance on randomly selected dates, either through pop-up quizzes or in-class questions. You are allowed to miss one class when attendance is taken. After that, each missing attendance will result in a deduction of two points, capped at fifteen points in total. Late arrivals will result in deduction of points according to degree of severity. Use of unauthorized devices will result in a deduction of no less than two points.

I will observe your behavior in the classroom. Being late to class or disruptive to the learning environment will result in deductions of participation points.

GRADING

Assignments	Points [or percentage]
Midterm	40%
Presentation/Write-up	50%
Participation	10%
Total	100%

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.

RESPECT FOR DIVERSITY

I intend to present the content of this course in a manner that is respectful of diversity: gender, sexuality, disability, age, socioeconomic status, ethnicity, race, religion and/or culture. Students are encouraged and expected to participate in class but are also expected to be respectful of their peers' diverse backgrounds and religious beliefs. Please let me know ways to improve the environment in the class for you personally or for other students or student groups. Please let me know:

- If you feel like your performance in the class is being impacted by your experiences outside of class.
- If you have a name and/or set of pronouns that differ from those that appear in your official records.
- If something was said in class (by me or anyone) that made you feel uncomfortable.